Poster: GEM-PLI Side Channel in GPON: Inferring Encrypted Streaming Traffic

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***Abstract*—Gigabit Passive Optical Networks (GPON), an ITU-T standardized technology adopted in FTTH networks, supports downstream payload encryption to protect subscribers' traffic. Unfortunately, encryption does not cover all access layer metadata. This poster presents a preliminary analysis demonstrating that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, can be exploited as a side channel to infer streaming activity. Using a GPON downstream simulator, We demonstrate that PLI traces are highly consistent across repeated runs of the same Youtube video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim's streaming activity (e.g., Youtube Video Identification).**

I. Introduction And Motivation

As fiber internet has become dominant in network infrastructures, GPON–one of the passive optical network (PON) technologies–is adopted by many ISPs to provide fiber-to-the-home (FTTH) internet. Since GPON is a popular PON technology, it is important to ensure traffic confidentiality. GPON supports AES-based downstream payload encryption at the GPON Encapsulation Method (GEM) layer. However, some fields in the GEM header remain plaintext such as the Payload Length Indicator (PLI) field used for multiplexing and the Port-ID field used for mapping subscribers' Optical Network Terminals/Units(ONTs/ONUs) (typically called "modem").

Such plaintext fields may appear innocuous. However, recent studies have demonstrated that such metadata may be exploited to infer a victim's activity (e.g., identifying a victim's web-based video streaming watching activity) [4] [5] [6].

While prior studies primarily focused on upper layer or cellular traffic analysis [4] [5], we researched whether the plaintext PLI field in GEM headers can be exploited to infer a victim's streaming activity.

In this poster, We developed a software-based GPON downstream simulator to analyze the statistical properties of PLI traces. Using these PLI traces, we present preliminary evidence that Youtube streaming traces are highly consistent over time and distinctive at the GEM layer, suggesting that value of the PLI field can be exploited to infer victim's video streaming.

II. Background

GPON framing. In GPON, the Optical Line Terminal (OLT) broadcasts downstream traffic over a passive optical splitter to multiple Optical Network Units (ONUs) (typically called "modem"). Downstream transmission is organized into fixed duration GTC frames (125 μs), each composed of a physical control block downstream (PCBd) followed by a downstream GTC payload region. [1]

GEM and plaintext metadata. Upper layer traffic (e.g., Ethernet frame) is carried in the Payload Field in GEM frames. A GEM frame starts with a 5 byte GEM header containing a 12 bit PLI, a 12 bit Port-ID, a 3 bit Payload Type Indicator (PTI), and a header error control (HEC) field. PTI indicates whether the payload is the end of a fragmented payload or not. [1]

PLI length even under encryption. In GPON, downstream user traffic is carried as GEM frames multiplexed in the GTC payload [1]. Each GEM header includes a plaintext PLI field that specifies the payload length L in bytes (up to 4095 bytes). In certain cases, larger payload values are fragmented accordingly [1](where PTI is used to indicate whether a GEM payload is a end of a fragment or not). Downstream GEM encryption only encrypts payload field(not GEM header) using AES-CTR. Since AES-CTR Encryption is length-preserving. even when payload bytes are encrypted, an attacker can observe the plaintext payload length.

Video Identification. HTTP Adaptive Streaming (HAS) is the popular video streaming model – which is generally adopted by video streaming platforms (e.g., Youtube) –delivers video by letting the client periodically request short media segments at a bitrate chosen by an adaptive bitrate (ABR) algorithm. This request–response process forms characteristic bursty download patterns and bitrate-dependent segment sizes, which can be exploited for traffic inference. [4]

III. Threat Model

We consider an attacker who is on the same optical splitter as the victim and has the capability to obtain the victim's Port-ID, which is crucial for targeting specific GEM frames.

- **1. Premise** The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT. the attacker does not need to register their device with OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting.
- **2. Collect GEMs** Then, attacker extracts GTC frames from downstream traffic and parses GEM frames.
- **3. Convert GEM PLI to dataset** Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps.

- **4. Dataset format** Each record includes `t_start_us`, `gem_id`, `port_id`, and `pli`.
- **5. Classification via database matching** The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

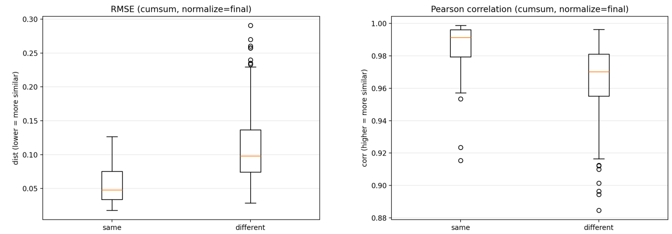
IV. Trace Generation

To demonstrate the threat model in our experimental environment, we developed a software based GPON downstream trace generator [3] that follows key ITU-T G.984.3 TC-layer parameters relevant to PLI (125 μs framing and GEM header formatting). From a downlink ethernet packet capture file of encrypted video traffic, we (i) map Ethernet frames into GEM payloads (fragmenting to satisfy the 12 bit PLI limit), (ii) generate plaintext GEM headers (PLI/Port-ID/PTI), and (iii) pack the GEM stream into fixed-length 125 μs downstream frames consistent with the configured line rate. The attacker signal is either the GEM-level PLI sequence or the per-frame occupancy series. Large Ethernet packets may generate multiple GEM fragments; thus, PLI values in our traces reflect per fragment payload lengths (the attacker-visible observable).

Note. Payload bytes can optionally be AES-CTR protected; however, this does not change the PLI trace because AES-CTR is length-preserving (XOR), so encryption does not change any fragment length. Our analysis uses only plaintext headers (e.g., PLI) and timing.

V. Experiment and Preliminary Results

First, we played each of 10 videos (three runs each; run1, run2, run3) on Ubuntu 24.04 under a fixed playback quality and resolution setting (720p). To minimize interference, we restricted other background traffic (e.g., VNC, SSH) during playback. For each run, we captured downlink ethernet packet (PCAP) for 180 seconds. Then, we replayed each PCAP file in our simulator to produce a CSV dataset of PLI values and timestamps(one trace per run) (e.g., vid001_run001.gem.csv [3]). We preserve original PCAP packet timestamps to retain idle periods and burstiness; each synthesized GEM is assigned t_start_us derived from the PCAP time base (µs resolution). After producing the dataset, we visualized cumulative PLI–Time Plot based on our dataset(Publicly Available on our Github repository) [3]. We observed a clear correlation between each video and its cumulative PLI over time graph [3]; traces from the same video show highly consistent slopes across runs.(Our Github repository: /out_cumsum) For similarity analysis, we bin PLI events into 100ms bins, compute the cumulative sum, normalize by the final cumulative value, and compare traces using Pearson correlation and RMSE. These traces approximate the payload length observations available to a real same-splitter adversary and enable systematic evaluation of PLI based inference. We use run1,2 dataset files as a database and run3 dataset files as a query trace.



(a) RMSE (lower = more similar) (b) Pearson correlation (higher = more similar)

Fig. 1. Similarity distributions between same-video and different-video cumulative PLI traces. Same video pairs tend to show higher correlation and RMSE smaller distance. We found a partial intersection due to ABR variability and buffering, but the distinction is sufficient for discrimination using both RMSE and Pearson

TABLE I
Per-query Top-1 predictions. [3]

Query	**Prediction**	**CORRECT**	**margin**	**r/RMSE**
vid001/r3	vid001/r1	T	2	0.998/0.025
vid002/r3	vid002/r1	T	5	0.995/0.041
vid003/r3	vid003/r2	T	1	0.980/0.050
vid004/r3	vid004/r1	T	2	0.996/0.037
vid005/r3	vid005/r2	T	2	0.998/0.020
vid006/r3	vid008/r1	F	2	0.994/0.048
vid007/r3	vid007/r1	T	2	0.999/0.025
vid008/r3	vid008/r1	T	2	0.999/0.018
vid009/r3	vid009/r2	T	2	0.990/0.041
vid010/r3	vid010/r1	T	3	0.998/0.020

We quantify similarity on the normalized cumulative PLI time series: same video pairs exhibit higher Pearson correlation (median $r = 0.991$ vs 0.970, AUC=0.779) and lower RMSE (median 0.048 vs 0.098, AUC=0.826). Using nearest matching with rank voting over Pearson and RMSE, we achieve 90% Top-1 identification accuracy (9/10) when matching run3 queries against run1,run2 query traces.

VI. Conclusion And Discussion

Despite GEM Payload Encryption, GPON may remain vulnerable to streaming inference because PLI values in GEM headers can be exploited to infer a victim's video activity (e.g., Youtube). Potential mitigations include traffic shaping and obfuscating GEM metadata.

Future Work will be conducted in testbed with Real OLT, ONU and Optical Splitter. We will use FPGA device to capture GEM frames to parse downstream frames. We will expand to more videos and longer captures with changing various resolution, bitrate.

Extent to XGS-PON Networks. We assume that XGS-PON has same issue; XGEM Frame that XGS-PON uses has Payload Length Indicator Field like GEM. We will validate this by repeating our capture-and-matching evaluation on an XGS-PON testbed.

References

[1] ITU-T Recommendation G.984.3, *Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification*.

[2] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230)

[3] Our Github Repository - https://gitshare.me/repo/c8afd178-a072-4cee-a01c-8b3d12467a28

[4] S. Bae, et al., "Watching the Watchers: Practical Video Identification Attack in LTE Networks," in *31st USENIX Security Symposium (USENIX Security '22)*, 2022.

[5] X. Zhang, et al., "Traffic spills the beans: A robust video identification attack against YouTube," *Computers & Security*, vol. 137, 2024, Art. 103623.

[6] N. Hu et al., "Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features," in *ESORICS 2024*, 2024.

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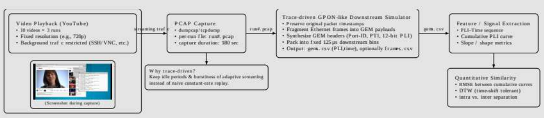
Title: Protocol vulnerability in Gigabit Passive Optical Networks (GPON)
This poster presents a preliminary analysis showing that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, may be exploited as a side channel that enables inference of streaming activity. Using a GPON downstream simulator, We demonstrate that PLI traces are highly consistent across repeated runs of the same video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim's streaming activity (e.g., Youtube Video Identification)

INTRODUCTION

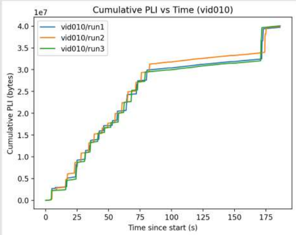
As FTTH(Fiber-To-The-Home) Internet has became popular, it is important to ensure traffic confidentiality. GPON(Gigabit Passive Optical Networks) supports AES based downstream traffic encryption but some fields in GEM(GPON Encapsulation Method) frame remain Plaintext.
You may think such fields is not important.
However recent works demonstrated that such fields can be exploited to inter user's video streaming activity(e.g., Leaking the name of the video user is watching)
In this poster, We Present preliminary evidence that PLI values in GEM header can be exploited to inter a victim's video activity.

EXPERIMENT

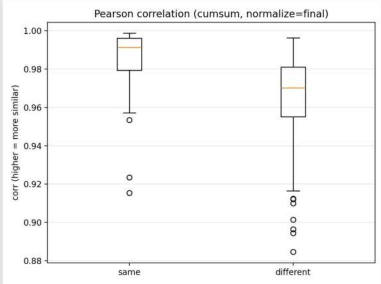


1. we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets.
2. At first we played each of 10 videos three times(RUN1, RUN2, RUN3) in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference.
3. Then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces dataset.
4. After producing dataset we visualized cumulative PLI - Time Graph(Please Look at right side)
5. Also We quantified similarity on the normalized cumulative PLI time series(Using nearest matching over Pearson and RMSE)

EXPERIMENT RESULTS



Cumulative PLI - Time Grapgh for one video showing that greatly identical slopes between three runs, PLI - Time Grapgh



Pearson correlation (higher=more similar)
AUC=0.779

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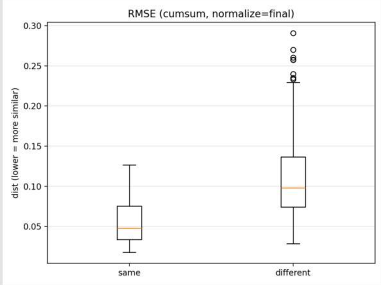
vid005/r3	vid005/r2	T	2	0.998/0.020
vid006/r3	vid008/r1	F	2	0.994/0.048
vid007/r3	vid007/r1	T	2	0.999/0.025
vid008/r3	vid008/r1	T	2	0.999/0.018
vid009/r3	vid009/r2	T	2	0.990/0.041
vid010/r3	vid010/r1	T	3	0.998/0.020



RMSE (lower=more similar)
AUC=0.826

*In rank-vote fusion of Perason and RMSE, We Achieve 90% Top-1 Video Identfication (*Query: Answer, Prediction: Predicted value obtained through RMSE and Pearson correlation., CORRECT: Is that Correct Prediction(True/False)*
r1=RUN1
r2=RUN2
r3=RUN3
We used RUN3 as query and RUN1,2 as database.

These results suggest that encrypting payload bytes alone is insufficient for privacy in GPON, motivating mitigations such as traffic shaping and metadata obfuscation.

CONTRIBUTIONS

- *access-layer side channel in encrypted GPON: We identify and empirically study the GEM-header PLI field (plaintext) as a threat source that can enable streaming inference even when GEM payloads are AES CTR encrypted.*
- *E2E attacker observation pipeline: We formalize a same-splitter adversary who (i) passively captures downstream traffic, (ii) parses GTC/GEM frames, (iii) filters by the victim's Port-ID, and (iv) obtains timestamped PLI traces for database matching–based video identification.*
- *Reproducible trace generation via a GPON downstream simulator: We build a software trace generator that follows key ITU-T G.984.3 TC-layer parameters relevant to PLI (e.g., 125 µs framing, GEM header formatting, and fragmentation under the 12-bit PLI limit) to synthesize attacker-visible PLI sequences from encrypted downlink PCAPs.*
- *Evidence of temporal consistency and distinctiveness at the GEM layer: Across repeated runs of the same YouTube video, cumulative PLI over time traces exhibit consistent slopes, while different videos tend to be separable under similarity metrics (Pearson correlation / RMSE).*
- *Preliminary video identification results: Using nearest-neighbor database matching with rank voting over Pearson and RMSE, we achieve 90% Top-1 accuracy (9/10) when matching run3 queries against run1,run2 traces on 10 videos.*

FURUTE WORK & LIMITATION

- *Simulator-driven results → hardware validation needed.*

Our current evidence is obtained from a software GPON downstream trace generator driven by encrypted downlink PCAP replays. As next steps, we will validate the attack end to end on a real GPON testbed (Real OLT/ONU and an optical splitter) using an FPGA based optical capture setup to parse downstream GTC and GEM frames.

- *Port-ID assumption and field feasibility.*

We assume the attacker can obtain and reliably track the victim's Port-ID to filter GEM frames. Future work will study practical Port-ID robustness in the field (e.g., stability across time, vendors, and operational conditions).

- *Scale and diversity of evaluation.*

The current dataset is limited (10 videos, three runs each, fixed 720p, 180s captures). We will expand to more videos and longer sessions, and systematically vary resolution,bitrate,ABR behavior to quantify how identification accuracy degrades or improves under realistic streaming dynamics.

- *Real splitter confounders not fully captured.*

A real PON may include cross-subscriber multiplexing noise, OMCI, and vendor's scheduling/packing artifacts. Future measurements on a shared splitter will characterize these confounders and test mitigation sensitivity.

REFERENCES

[1] ITU-T Recommendation G.984.3, Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification.

[2] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230)

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Our Simulator's Code and dataset are publicly available in our Github repository.